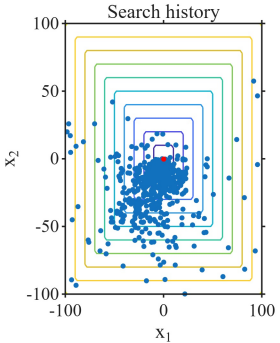
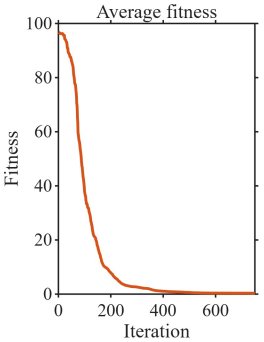
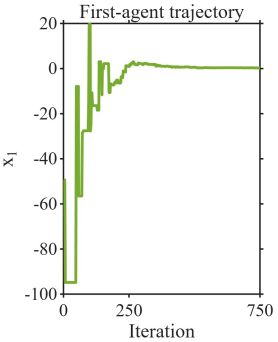
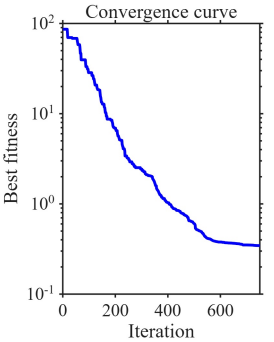

F4
Fitness
100
50
0
100
0
-100
x2
-100
0
100
x1


Search history
x2
100
50
0
-50
-100
-100
0
100
x1


Average fitness
Fitness
100
80
60
40
20
0
0
200
400
600
Iteration


First-agent trajectory
x1
20
0
-20
-40
-60
-80
-100
0
250
750
Iteration


Convergence curve
Best fitness
10^2
10^1
10^0
10^-1
0
200
400
600
Iteration